Urban Waste Management:

Policy & Implementation Assessment

INDEPENDENT RESEARCH | 2026

MANYA MISHRA

Table of Contents

EXECUTIVE SUMMARY

In this era of globalization and fast consumerism waste generation has accelerated. Per capita waste generation has increased significantly from 2015 to 2025. The current situation is demanding more change in the waste management practices than before. The overall quantity of solid waste generated stands at 160,038.9 tons per day (TPD) in the country and the hazardous waste generation increased significantly from 10.92 MMT in 2020–21 to 19.02 MMT in 2024–25. Inadequate handling of Municipal Solid Waste (MSW) poses significant challenges, especially concerning its influence on the expanding population.

Solid waste management needs to be reshaped by adopting a holistic approach and inculcating a culture of accountability. This transformation requires shifting away from reliance on landfills/open dumps and transitioning towards a circular economy model, where waste is viewed as a valuable resource.

Municipal budgets are under pressure, primarily allocated towards employee salaries, leaving limited resources for investing in SWM infrastructure. To tackle these issues, it is crucial to explore financial strategies like increasing property taxes, implementing feasible and viable business models for waste treatment such as composting and waste-to-energy facilities, and engaging the private sector to boost funding for SWM projects.

These measures can enhance the financial sustainability of SWM initiatives and contribute to better waste management practices across the country. Integrating private sector involvement will effectively complement state efforts, given the finite or stagnating capability of governmental resources over time. There is no one-size-fits all approach to solving the complex issue of solid waste management; instead, a comprehensive plan is necessary.

1) POLICY PROBLEM

Waste is often neglected because of it less to no use and often the waste management problem is overlooked with the same notion. On the contrary not managing waste could lead to much worse problem and though its repercussions are delayed they are much more impactful. Problem of urban waste starts from heavy waste generation and ends at failure of this heavy waste management. The implementation of three R's i.e. Reduce, Reuse, and Recycle is not met. Though it is an old problem yet now both the quantity and area of this problem has increased. Now the urban cities are facing an increase in the influx of waste where are we can see the problem reaching the villages.

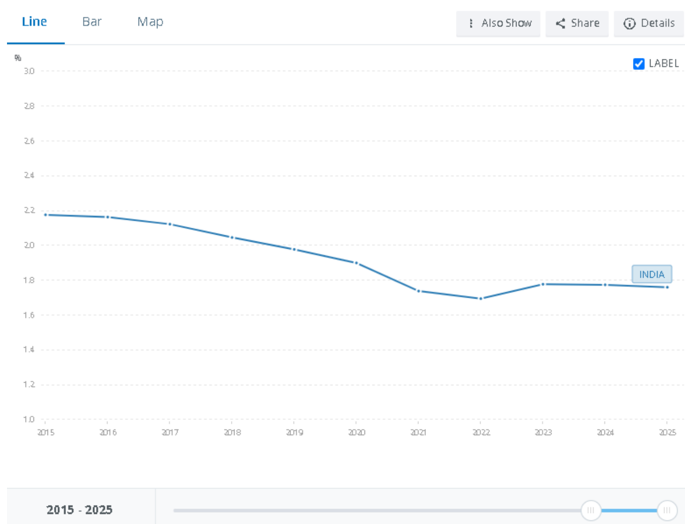
STAKEHOLDERS

- **Residents:** Lack of civic sense and awareness along with high ignorance makes the residents the major victim of waste as well as major generator.
- **Municipal Bodies:** Municipal bodies are the first executive body to act on the waste management: lack of staff, not having proper supervision on contractual waste collector, lack of monitoring at every stage of waste collection encourages further the issue created by residents
- **Waste workers:** Waste workers often don't follow the guidelines and usually leave a crucial step of segregating the waste and the end to end procedure of recycling if often not met.
- **State and Central Government**: The lack of data and lack of policy implementation on ground often comes as a bigger road block. Adding to that there is not schemes or solution for the already accumulated waste.

2) BACKGROUND & URBAN WASTE LANDSCAPE IN INDIA

2.1 Urban population growth

According to World Bank data growth rate of urban population of India is currently 1.8 in 2025 declined from 2.2 in 2015. This shows that the issue with waste is not increase in population but same number of people are producing more amount of waste hence the issue is consumerism,



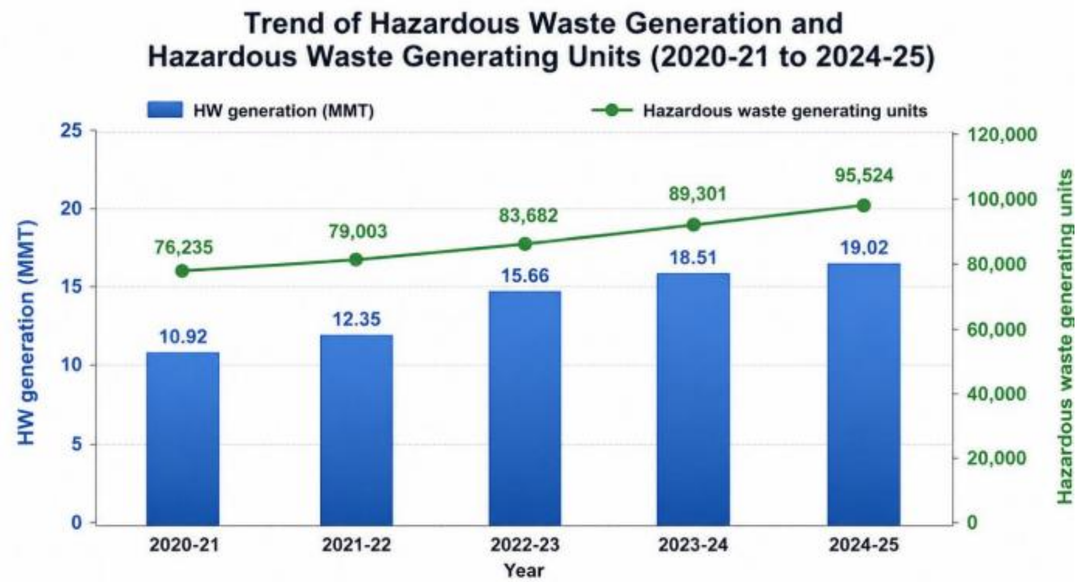
2.2 Waste generation trends

According to the CPCB (Central Pollution Control Board) report 2020-21, the overall quantity of solid waste generated stands at 160,038.9 tons per day (TPD) in the country. Out of which 152,749.5 TPD of waste is collected in an efficient manner. Out of the total collected waste, 79,956.3 TPD, constituting 50 per cent, undergoes some form of treatment, while 29,427.2 TPD, i.e., 18.4 per cent, is directed to landfills and 50,655.4 TPD, representing 31.7 per cent of the total waste generated remains unaccounted

SOLID WASTE GENERATION PER CAPITA

Year	Solid Waste Generation Per Capita(gm/day)
2015-16	118.68
2016-17	132.78
2017-18	98.79
2018-19	121.54
2019-20	119.26
2020-21	119.07

TRENDS OF HAZARDOUS WASTE GENERATION (2020-21 TO 2024-250



Year	2020-21	2021-22	2022-23	2023-24	2024-25
HW generation (MMT)	10.92	12.35	15.66	18.51	19.02
Hazardous waste generating units	76,235	79,003	83,682	89,301	95,524

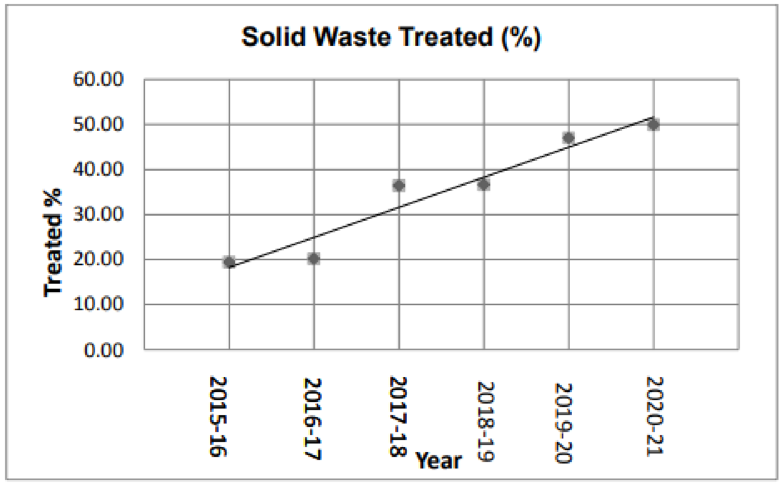
The graphical analysis indicates a steady increasing trend in both hazardous waste generation and the number of hazardous waste-generating units across the country during the period 2020–21 to 2024–25. Hazardous waste generation increased significantly from 10.92 MMT in 2020–21 to 19.02 MMT in 2024–25, reflecting an overall growth of about 74% over the five year period. Similarly, the number of hazardous waste-generating units increased from 76,235 to 95,524, indicating continued expansion of industrial and manufacturing activities. The consistent rise in both parameters highlights the need for strengthening hazardous waste management infrastructure, including recycling, utilization, treatment, and disposal facilities, to ensure environmentally sound management of the increasing waste quantities.

2.3 Collection rates

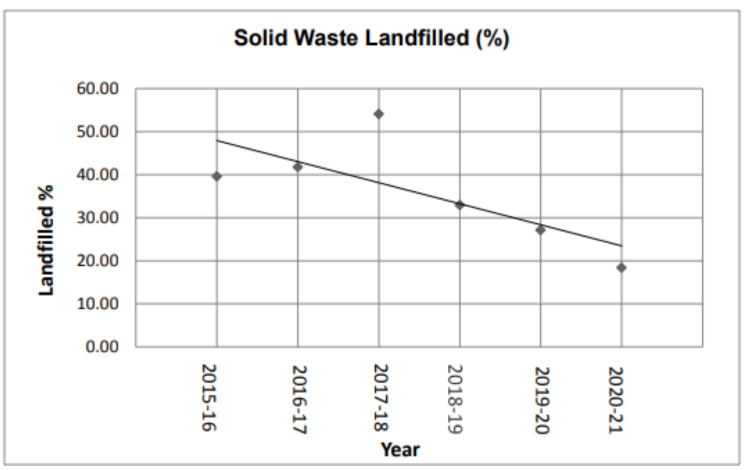
In context of domestic waste only 15 SPCBs/PCCs namely Bihar, Chandigarh, Chhattisgarh, Goa, Haryana, Kerala, Karnataka, Madhya Pradesh, Puducherry, Punjab, Rajasthan, Tamil Nadu, Telangana, Uttar Pradesh & West Bengal have reported other waste recyclers in their States/UTs. There are 650 other waste recyclers operating in these States/UTs. Out of which, Karnataka (218) has maximum number of recyclers followed by Tamil Nadu (151). Out of 650 recyclers, 312 units are authorized for Schedule III Part B other wastes having authorized capacity of 8.45 Million MT and 338 units for Schedule III Part D having authorized capacity of 9.62 Million MT. During 2024-25, about 5.72 Million MT of other wastes (imported/domestically) was recycled/utilized, the table showing the data is provided below:

S.No.	States/UTs	No. of units authorized for recycling/ utlization of Other Waste (MT)		Authorized Capacity (MT)		Quantity of other waste imported from other country (MT)	Quantity of other waste exported to other country (MT)	Quantity of other waste (Schedule III waste B and D) utilized/recycled during the year April-March (MT)		Other waste sent for disposal to Common TSDF (MT)	Quantity of HW generated during recycling/ utilization of other waste (MT)	Quantity of other waste stored at Occupiers premises (MT)	
		Schedule III-Part B	Schedule III-Part D	Schedule III-Part B	Schedule III-Part D			Imported	Domestically generated			Beginning of the financial	End of financial year
1.	Bihar	0	2	0	27000	1313	0	1313	14	0	2	47	12
2.	Chandigarh	0	3	0	12400	20	0	20	282	0	0	52	16
3.	Chhattisgarh	49	6	1646660	789841	14319	0	11716	330134	0	0	26720	13703
4.	Goa	0	12	0	996900	28146	0	28146	121364	0	0	0	0
5.	Haryana	69	5	546129	459000	14926	0	14926	447219	34	113	1101	4255
6.	Kerala	5	20	0	1925172	13276	0	12222	256387	0	0	205	419
7.	Karnataka	95	123	390300	330693	68951	4252	66906	199803	467	5082	5934.1	8399
8.	Madhya Pradesh	25	0	261689	0	0	0	0	6780	8744	0	0	0
9.	Puducherry	0	10	0	409270	124183	0	119178	172927	0	0	14716	29734
10.	Punjab	19	0	32212	0	2266	0	10479	42422	0	0	0	0
11.	Rajasthan	24	14	5022290	91890	29502	0	29490	1450454	0	20	103121	82369
12.	Tamil Nadu	14	137	249113	4542215	651782	0	634036	1459761	0	803	85298	107798
13.	Telangana	2	2	2751	1012	0	0	0	786	0	0	0	0
14.	Uttar Pradesh	10	0	294617	0	294617	0	294617	0	0	0	0	0
15.	West Bengal	0	4	0	35180	5334	0	5334	0	0	0	0	0
	Total	**312**	**338**	**8445761**	**9620573**	**1248636**	**4252**	**1228383**	**4488333**	**9245**	**6019**	**237194**	**246704**

2.4 Processing/ Recycling



While agents of government are present for waste processing and recycling people contributing to that are rather less. People participation in waste segregation is often less. For people actively participating in waste segregation the waste not always ends up at recyclers rather dumped on grounds. Yet we can see increasing trend in percentage of solid waste processed has been observed during the last five years wherein percentage of solid waste processed has increased from 19% in 2015-16 to 49.96% in 2020-21. In regards with hazardous waste there are 2448 authorized recyclers having authorized capacity of 12.68 Million MT. The state of Maharashtra (628), Gujarat (354), Haryana (206), Uttar Pradesh (201), Karnataka (151), Madhya Pradesh (134), & Rajasthan (123) have major number of recyclers, followed by West Bengal (121). During the year, about 2.51 Million MT of hazardous waste has been recycled.



2.5 Landfill dependency

A solid waste landfill is a carefully engineered site designed to dispose of non-hazardous garbage safety. A decreasing trend in solid waste landfilled has been observed during the last six years wherein solid waste landfilled has decreased from 54% in 2015-16 to 18.4% in 2020-21 siting that the other methods of waste disposal are not being actively used.

3) EXISTING POLICY & LEGAL FRAMEWORK

3.1 Central framework

3.1.1 Environment (Protection) Act, 1986

This Act provide for the protection and improvement of environment and for matters connected therewith. It is an umbrella legislation enacted by the Parliament of India. Passed in May 1986 in the wake of the Bhopal gas tragedy, it came into force on November 19, 1986. It grants the central government broad powers to protect and improve the human and natural environment.

3.1.2 Solid Waste Management Rules, 2016

The Solid Waste Management Rules, 2016 were notified by India's Ministry of Environment, Forest and Climate Change to overhaul municipal waste handling. It mandated waste generators to sort waste at source into biodegradable (wet), dry (recyclable), and domestic hazardous waste. It prohibited the burning, burying, or random littering of municipal solid waste and encouraged brand owners and manufacturers of-packaged goods to assist local bodies in waste collection systems.

3.1.3 Plastic Waste Management Rules, 2016

The Plastic Waste Management Rules, 2016 aim to: Increase minimum thickness of plastic carry bags from 40 to 50 microns and stipulate minimum thickness of 50 micron for plastic sheets also to facilitate collection and recycle of plastic waste, Expand the jurisdiction of applicability from the municipal area to rural areas, because plastic has reached rural areas also; To bring in the responsibilities of producers and generators, both in plastic waste management system and to introduce collect back system of plastic waste by the producers/brand owners, as per extended producers responsibility; To introduce collection of plastic waste management fee through pre-registration of the producers, importers of plastic carry bags/multilayered packaging and vendors selling the same for establishing the waste management system.

3.1.4 Construction & Demolition Waste Management Rules, 2016

The Government has notified Construction & Demolition Waste Management Rules, 2016 for the first time. Outlining the salient features of the Construction & Waste Management Rules they are an initiative to effectively tackle the issues of pollution and waste management. In 2016, the construction & demolition waste generated is about 530 million tonnes annually this construction & demolition waste is not a waste, but a resource. The basis of these Rules is to recover, recycle and reuse the waste generated through construction and demolition. The segregating construction and demolition waste and depositing it to the collection centres for processing will now be the responsibility of every waste generator.

3.1.5 E-Waste Management Rules, 2022

These rules intend to manage e-waste in an environmentally sound manner and put in place an improved Extended Producer Responsibility (EPR) regime for e-waste recycling wherein all the manufacturer, producer, refurbisher and recycler are required to register on portal developed by CPCB. The new provisions would facilitate and channelize the informal sector to formal sector for doing business and ensure recycling of E-waste in environmentally sound manner. Provisions for environmental compensation and verification & audit has also been introduced. These rules also promote Circular Economy through EPR regime and scientific recycling/disposal of the e-waste.

3.1.6 National Green Tribunal

In the last five years July 2018 to July 2023 several innovative steps have been taken by National Green Tribunal with a focus to simplify the procedures to make National Green Tribunal people friendly and to improve faster disposal of matters including: Various initiatives relating to court management and case management were taken such as identifying issue and scope of proceedings in first order. Zero adjournments were given which resulted in faster disposal of cases thereby leading to disposal of numerous matters relating to national importance. Joint committees were constituted which were often headed by retired Judges and statutory regulators which enabled independent ascertainment of factual position leading to speedy disposal. Technology was used to increase efficiency by way of service of notice through email, all filings in soft copy, placing of reports on website. Other than that Suo Motu interventions to remedy environmental degradation manifested by online data maintained by statutory regulators and Suo motu interventions for compensation and protection of environment in cases of fatal accidents involving violation of environment safety norms which helped common man get expeditious relief on the basis of principle of restitution.

3.2 Institutional responsibilities

3.2.1 MoEFCC

The Ministry of Environment, Forest and Climate Change notified the Manufacture, Storage and Import of Hazardous Chemicals (MSIHC) Rules, 1989 and the Chemical Accidents (Emergency Planning, Preparedness and Response) (CAEPPR) Rules, 1996 for ensuring chemical safety in the Country. These rules delineate the criteria for identification of Major Accident Hazard (MAH) unit. As per the rules, Central Crisis Group, State Crisis Groups, District Crisis Groups and Local Crisis Groups at Central, State, District and Local level are required to be set up for the management of accidents due to handling of hazardous chemicals listed in the rules. An off-site emergency plan for a district having MAH unit(s) is required to be in place so as to mitigate the impact of chemical accidents. As per the information received from various States and Union Territories, there are 1,861 MAH units in the Country, located in 303 districts.A sub-scheme titled "Industrial Pocket wise Hazard Analysis" has been in operation since the Eighth Five Year Plan. The Ministry provides financial assistance for preparation of off-site emergency plans, hazardous analysis and rapid safety audit reports to identified agencies for preparation of Off-Site Emergency Plans for 41 districts in the country having MAH units. Reports have been received and these off-site emergency plans including hazard analysis and rapid safety reports are under review.

3.2.2 CPCB

Advise the Central Government on any matter concerning prevention and control of water and air pollution and improvement of the quality of air. Plan and cause to be executed a nation-wide programm for the prevention, control or abatement of water and air pollution; Co-ordinate the activities of the State Board and resolve disputes among them; Provides technical assistance and guidance to the State Boards, carry out and sponsor investigation and research relating to problems of water and air pollution, and for their prevention, control or abatement. Advises the Governments of Union Territories with respect to the suitability of any premises or location for carrying on any industry which is likely to pollute a stream or well or cause air pollution. Lays down standards for treatment of sewage and trade effluents and for emissions from automobiles, industrial plants, and any other polluting source. Evolve efficient methods for disposal of sewage and trade effluents on land; develop reliable and economically viable methods of treatment of sewage, trade effluent and air pollution control equipment; Identify any area or areas within Union Territories as air pollution control area or areas to be notified under the Air (Prevention and Control of Pollution) Act, 1981.

3.2.3 State Pollution Control Boards

The State Pollution Control Board for Prevention and Control of Water Pollution was constituted by the Government of each state in pursuance of the Water (Prevention & Control of Pollution) Act, 1974. The Water Act provides for the prevention and control of water pollution and the maintaining or restoring of the wholesomeness of water. After the enactment of the Air (Prevention & Control of Pollution) Act, 1981, the enforcing responsibility was entrusted to the above Board. The Air (Prevention & Control of Pollution) Act, 1981 is an enactment to provide for the prevention, control, and abatement of air pollution. Apart from the above said Acts, the Board is also enforces the rules and notifications framed under the Environment (Protection) Act, 1986

4) CURRENT SITUATION: Where did the system break down?

A significant roadblock to the effective management of solid waste is the process of waste segregation. Before the waste is collected from sources, segregation makes the process efficient, further facilitating the recycling of the waste and reducing the cost of waste disposal. As per literature, most households (especially the households of underserved communities such as slums, etc.) and other establishments dump garbage in open spaces, drains and water bodies, and inappropriate places. Due to the absence of waste segregation, recycling wastes has been associated with the informal sector through old-fashioned technology; however, this also leads to the production and demand of cheap and affordable products in the market. Paper and plastics are some common examples of these products. Several developed nations, such as France, Austria, etc., have formulated laws and regulations to enforce mass recycling of solid waste to cope with waste generation on a daily basis.

Waste collection, storage, and transportation are essential parts of any solid waste management system, and the implementation of these tasks can be quite challenging. As per Municipal Solid Waste Management Rules 2016, there are two kinds of waste collection methods, namely primary and secondary collection. The former involves the waste collected from the source it is generated, while the latter refers to the waste collected from sources such as community bins, storage, etc., to the processing or disposal location. In India, waste is collected formally and informally due to various factors, such as inadequate finance, lack of efficient technology, shortage of trained workers, etc. Balasubramanian (2018), in his study, highlighted that the schemes of solid waste management collection benefit only some sections of the urban population because the available resources and capacity of planning are not able to cope with increasing waste generation.

Disposal and treatment of solid waste management is a critical issue in the development and growth of human settlements worldwide. Merely discarding solid waste out of sight does not resolve the problem; instead, it often exacerbates the problem, leading to a multitude of adverse consequences. Over time, this unchecked disposal can spiral out of control, posing serious challenges for everyone involved. In India, a staggering majority, exceeding 90 per cent, of municipal solid waste in urban areas is disposed of directly onto land in unsatisfactory conditions.

5) KEY POLICY GAPS

Policy Area	Existing Provision	Solution	Policy Gap
Inadequate Municipal Finance	Stagnation of municipal revenues and expenditure	Private Investment	Inadequate finances
Technology & work force	Lack of workforce	Capacity building	Capacity building of municipal workers
Structural Strategie	SWM projects	Composting	Intervention to improve solid waste management practices
Municipal Finance	Stagnant financing	Alternative funding source	Tax financing
Structural Strategie	Lack of waste management	Waste to Energy plants	Efficient waste treatment

6) COMPARITIVE CASE STUDIES

Japan as a whole generated approximately 40.34 million tonnes of municipal solid waste in FY 2022, equivalent to around 110,600 tonnes per day, which is significantly lower than India's 160,000 tonnes per day despite comparable scales of urban consumption and GDP. Systematic segregation, Extended Producer Responsibility (EPR) compliance and advanced processing ensure that landfills account for less than 5% of disposal. Japan demonstrates that the same waste quantities can lead to very different outcomes depending on how the value chain of solid waste management is governed.

A stage-wise comparison highlights India's bottlenecks and Japan's solutions:

Generation: In India, rapid urbanisation and consumerism drive rising waste volumes, but segregation at source is weak and EPR compliance remains poor. In Japan, waste generation is tightly controlled: producers bear extended responsibility, while households must follow detailed segregation guidelines across 10+ categories, enforced by penalties. India must treat segregation and EPR enforcement as the foundation of waste governance.

Collection & Transport: India collects 90- 95% of waste, but most is mixed. Trucks rarely use GPS or monitoring, limiting efficiency. Japan has 100% coverage, IoT-enabled trucks and route optimisation, supported by "Pay-As-You-Throw" (PAYT) systems where citizens pay for each prepaid bag of waste. This reduces volumes at source and improves accountability.

Segregation & Sorting: India achieves less than 60% segregation, with the informal sector salvaging 25% of recyclables. Japan exceeds 90% segregation, embedding sorting as a cultural norm. Smart bins by NEC and Panasonic use sensors and AI to guide users, while India's pilots (e.g., Pune, Hyderabad with firms like Vighnaharta and Convexicon) remain limited to collection efficiency. India must make segregation a citizen practice through incentives and penalties.

Processing & Treatment: India has composting and biogas pilots, plus informal recycling hubs, but lacks scale. Japan has pioneered Eco-Towns (Kitakyushu, Kawasaki), community-level Takakura composting and industrial loops where waste becomes input for another industry. India needs regulated recycling clusters and compost hubs integrated into municipal planning.

Residuals: India relies on hazardous dumpsites like Ghazipur and Deonar, with biomining promoted under SBM 2.0 but progressing slowly. Japan minimizes residuals through the Fukuoka semi-aerobic landfill method, which reduces methane, improves leachate quality and stabilises sites at costs comparable to Indian landfills (₹570/tonne vs ₹500-800/tonne) (Hanashima, 2002). India should combine biomining for legacy waste with Fukuoka-style engineered landfills.

Waste-to-Energy (WtE) & Disposal: India has WtE plants in cities like Delhi, Mumbai, Jabalpur but they face emission violations and fly-ash disposal problems. Japan incinerates >70% of MSW in robust stoker-type plants designed for low-calorific Asian waste, recovering energy for district heating with advanced flue-gas cleaning. Japanese firms like Hitachi Zosen already operate in India (Jabalpur, Pimpri-Chinchwad). India must enforce pre-sorting and strict emissions monitoring before scaling WtE.

This comparative value chain shows that while India has progressed in collection, it lags in segregation, processing and landfill design. Japan demonstrates that aligning policy, citizen compliance and technology at every stage transforms waste into value.

Toward a Localized Indo-Japan Framework

To bridge this gap, India must adapt, not copy, Japan's solid waste management system. A comparative techno-economic and policy framework can be built on three levers.

First, **technology transfer for scale**: modular composting units, IoT-based collection and mid-capacity incinerators can be adapted to India's mixed waste streams, while piloting Eco-Town clusters to integrate industry and informal workers.

Second, **investment models**: PPPs, Viability Gap Funding and concessional finance from JICA can de-risk projects and incentivize Japanese firms to bring advanced technologies.

Third, **manufacturing integration**: Local clusters producing WtE boilers, filters and recycling machinery under *Make in India* will reduce costs, create jobs and enable India to export circular economy technologies to South Asia and Africa.

If these interventions are adopted at a larger scale and implemented properly, they could divert 60 million tonnes of waste annually from landfills by achieving 80% household segregation (MoHUA, 2024). UNEP (2023) estimates this would cut 14 million tonnes of CO_2-equivalent methane emissions per year. CPCB (2024) projects creation of 150,000 direct jobs while recovering recyclables worth ₹2,500 crore annually. More broadly, localizing Japanese technologies would reduce landfill dependency, integrate informal workers and make India a regional hub for green technologies.

7) RECOMMENDATIONS & FINANCING MECHANISM

1. Actions on waste reduction to be undertaken in Mission Mode, as it involves multiple agencies and tasks, which have to act in sync with each other, with other projects and based on targets.
2. NITI Aayog SDG India Index, 2018, could incorporate waste reduction in a new section comprising SDG 12. This will spark competition as the baseline is set for each one.
3. Based on the measurement of waste reduction, states – both big and small - should be ranked and rewarded. Rewards can include fiscal means to further reduce waste, scaleup, etc.
4. Need to catalyze public opinion to reduce waste. Intensive campaign to encourage people to Have Less is essential. Using informal sector services for waste reduction should also be seen as desirable and on-trend, for widespread public acceptance.
5. Harmonise the Rules and Policies- Rules differ across states creating challenges for implementing agency or brand owner working or designing products for sale across

India. National harmonisation must reflect consistency on sizes and materials disallowed and undertaken by the MoEFCC.

6. Extended Producer Responsibility -EPR is already being used India for plastics (and Ewaste).
7. Existing EPR Mandate to be expanded in PWM Rules, 2016, Rules, and apply to MLPs, Tetrapak, Styrofoam, textiles with synthetic components. Must apply to brand owners, manufacturers and e-commerce. Laws notified within 9 months of year. Guidelines issued, with year wise targets to collect, recycle and report back.
8. To sustain ban by providing material alternatives and process alternatives and ensure continued supply. Investments in micro or medium industries with Government help in making them cheaper till scale reached. Process alternatives include Bartan Bhandars, or Crockery banks, to displace plastic crockery.
9. Every alternative is first verified by a state level task force, based on national guidelines. The task force at state level will comprise state government and agencies outside government, to verify the bio-degradability, safety and possible challenges of the material as a plastic alternative and issue a go-ahead. The task force can be state-wide, with representations in cities. All state task forces should be able to communicate with each other for mutual strengthening through a formal process.
10. Global experience suggests bans to be policed, awareness and strategic implementation important. Lessons for India for single use plastics: Implement bans phase-wise, with the easiest items first. Seek alternatives and advertise and promote them widely. Some single use plastics (SUPs) don't have alternatives, such as small sachets, and should be put in the final phase of the ban. Bans should be extended to all non-medical-use plastics and not merely restricted to plastic bags. In Phase 2, ban could be implemented on: Coffee cup lids, plastic plates, plastic and thermacol glasses, spoons, folks, knives, plastic lined glasses, PET water bottles of less than 200 ml. Adequate alternatives need to be ready. Wet wipes, flexi-banners.
11. Fiscal Instruments- can be used to discourage and create disincentives on use of plastics. Should be mandatory to charge minimum surcharge for selective items, till ban is completely effective. Diverse committee to identify fiscal measures for plastics, with a focus on nonrecyclable plastics. These can be at the level of brand owner and retail (straws, plastic bags, food containers etc.). The possible actor involved can be the National Institute of Public Finance and Policy (NIPFP).
12. Influencing Behaviour- Green Procurement: Government to consider green procurement as it procures for a range of uniforms, soft furnishings, etc. - to be mandated to be of natural fibres only. Positively impact pricing.

13. Map the range of actors and identify their perceived and other technological gaps. In some cases, technology can also be used for market access, such as in the case of service aggregators. Some of the technologies used are excellent examples of low-tech, high value. These include the cycle-pedal case of knife sharpeners. Some, like the cobblers, might need ergonomic interventions.
14. The service providers require skill training to transition into formal sector entrepreneurs and take advantage of several existing schemes, including loans etc. This should be provided to them in a format that is easy for them, including the number of days, timings, content and post-skilling hand-holding. Aggregating such persons at a city-wide level is key, for easy access

8) CONCLUSION

As human it is our responsibility to take care of our products, as we purchase the product from a definite place rather than picking anything from the ground and using it, it is our responsibility to treat the product in the same manner after we are done with it. Waste is nothing else but our responsibility as owner. If humans take accountability as agents of society then the three R's will be restored. Further industries needed to be regulated in high and tight manner by the government. If the policy for waste generation is made by government while keeping in mind that these are not just rules but a way to form a system that shall alter behavior of the citizens policy shall move more swiftly.

9) REFRENCES

- **NITI AAYOG: Final Report-Action Research for Waste Reduction:** https://www.niti.gov.in/sites/default/files/2023-03/Final%20Report-Action%20Research%20for%20Waste%20Reduction.pdf
- **PIB: Confronting the challenge of mounting waste (2017):** https://www.pib.gov.in/newsite/printrelease.aspx?relid=167191&reg=48&lang=2
- **CBCB: Annual Report 2024-2025:** https://cpcb.gov.in/openpdffile.php?id=UmVwb3J0RmlsZXMvMTg0MF8xNzc1NjQyNTY0X21lZGlhcGhvdG8yMjU3NS5wZGY=
- **World Bank Urban Population Growth:** https://data.worldbank.org/indicator/SP.URB.GROW?end=2025&locations=IN&start=2015&view=chart
- **Annual_Inventory2024-25:** https://cpcb.gov.in/uploads/hwmd/Annual_Inventory2024-25.pdf
- **CPCB (2021). Annual Report 2020-21 on Implementation of Solid Waste Management Rules, 2016:** https://cpcb.nic.in/uploads/MSW/MSW_AnnualReport_2020-21.pdf
- **EAC-PM Working Paper Series: Challenges of Solid Waste Management in Urban India**: https://eacpm.gov.in/wp-content/uploads/2024/05/Solid_Waste_management_Updated.pdf
- **PIB:** https://www.pib.gov.in/newsite/printrelease.aspx?relid=138144&reg=48&lang=2
- https://www.pib.gov.in/newsite/printrelease.aspx?relid=138389&reg=48&lang=2
- https://www.pib.gov.in/PressReleasePage.aspx?PRID=1986201&reg=48&lang=2
- https://www.pib.gov.in/PressReleasePage.aspx?PRID=1937745&reg=48&lang=2
- **MOEFCC:** https://moef.gov.in/waste-management
- https://www.iimb.ac.in/mijsc/kizuna-sabha/landfills-circular-cities.php